

Experiment 09: Input & Output Manual

Hosting the Website with Domain Registration Process

🎓 Student & Laboratory Credentials

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- **Academic Year:** 2026 – 2027
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- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Hosting the Website with Domain Registration Process

1. Experiment Aim

To analyze, document, and simulate the complete engineering workflow of procuring a custom Top-Level Domain (TLD) (`healthpulse-care.org`), delegating authoritative nameservers, structuring DNS resource records (A, CNAME, MX, TXT/SPF), and deploying production web applications across global edge cloud infrastructure (Vercel and Netlify).

2. 🖱️ Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
<code>vercel.json</code>	JSON	Vercel production edge deployment configuration: specifies clean URLs, security headers (<code>X-Frame-Options</code>), HSTS), and SPA rewrites.
<code>netlify.toml</code>	TOML	Netlify build pipeline configuration: declares build directory (<code>public</code>), redirect rules (<code>/* -> /index.html 200</code>), and static asset caching.
<code>package.json</code>	JSON	Project metadata and deployment scripts (<code>npm run deploy:vercel</code> , <code>npm run deploy:netlify</code>).
<code>public/index.html</code>	HTML5 / CSS3 / JS	Interactive DNS Zone Record Inspector and Propagation Terminal simulating CLI <code>dig</code> and <code>nslookup</code> queries.

2.2 Execution Command Input

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-09-Hosting-Domain-Process

# Option A: Open interactive DNS inspector directly in browser
chromium-browser public/index.html

# Option B: Run local preview server
npx serve public
```

2.3 Domain Registration & Nameserver Delegation Parameters

PARAMETER NAME	VALUE ASSIGNED	TECHNICAL PURPOSE
Domain Name	healthpulse-care.org	Fully Qualified Domain (FQDN)
Top-Level Domain (TLD)	.org (Public Interest Registry - PIR)	Healthcare & Clinical Portal
Accredited Registrar	Cloudflare Registrar / Namecheap	ICANN-accredited domain registry
Primary Nameserver	ns1.vercel-dns.com	Authoritative Anycast DNS Node 1
Secondary Nameserver	ns2.vercel-dns.com	Authoritative Anycast DNS Node 2
DNS Propagation TTL	3600 seconds (1 hour standard)	Local resolver cache expiry

2.4 Authoritative DNS Zone Resource Records Input Table

RECORD TYPE	HOST / NAME	TTL	DESTINATION / TARGET VALUE	OPERATIONAL FUNCTION
A Record	@ (Apex)	3600	76.76.21.21	Points apex to Vercel Edge
CNAME	www	3600	cname.vercel-dns.com	Canonical alias for www
MX Record	@ (Apex)	3600	10 mail.healthpulse-care.org	Priority 10 mail routing
TXT Record	@ (Apex)	3600	v=spf1 include:_spf.google.com ~all	Anti-spoofing SPF policy
TXT Record	_dmarc	3600	v=DMARC1; p=reject; rua=mailto:...	DMARC email authentication
NS Record	@ (Apex)	86400	ns1.vercel-dns.com	Authoritative NS delegate

2.5 Cloud Deployment Configuration Input Files

vercel.json :

```
{
  "version": 2,
  "cleanUrls": true,
  "trailingSlash": false,
  "headers": [
    {
      "source": "/(.*)",
      "headers": [
        { "key": "X-Content-Type-Options", "value": "nosniff" },
        { "key": "X-Frame-Options", "value": "SAMEORIGIN" },
        { "key": "Strict-Transport-Security", "value": "max-age=31536000; includeSubDomains" }
      ]
    }
  ],
  "rewrites": [
    { "source": "/(.*)", "destination": "/index.html" }
  ]
}
```

netlify.toml :

```
[build]
  publish = "public"
  command = "echo 'Building HealthPulse production assets...'"

[[redirects]]
  from = "/*"
  to = "/index.html"
  status = 200

[[headers]]
  for = "/*"
  [headers.values]
    X-Frame-Options = "SAMEORIGIN"
    X-Content-Type-Options = "nosniff"
```

3. Output Specification

3.1 Simulated DNS Resolution & Terminal Query Output ([dig](#) / [nslookup](#))

```
# Query A Record (IPv4 Address Resolution)
$ dig +noall +answer healthpulse-care.org A
healthpulse-care.org. 3600 IN A 76.76.21.21

# Query CNAME Record (Subdomain Canonical Alias)
$ dig +noall +answer www.healthpulse-care.org CNAME
www.healthpulse-care.org. 3600 IN CNAME cname.vercel-dns.com.

# Query MX Records (Mail Exchange Server Priority)
$ dig +noall +answer healthpulse-care.org MX
healthpulse-care.org. 3600 IN MX 10 mail.healthpulse-care.org.

# Query TXT Record (SPF & Domain Ownership Verification)
$ dig +noall +answer healthpulse-care.org TXT
healthpulse-care.org. 3600 IN TXT "v=spf1 include:_spf.google.com ~all"
healthpulse-care.org. 3600 IN TXT "google-site-verification=HP-992144-VERIFIED"
```

3.2 Graphical User Interface Output Representation ([public/index.html](#))

```
+-----+
| [HealthPulse Portal] SBL - Advanced Web Technology (Sem VI) Student: Asmit (25CE1051) |
+-----+
|
|      ▢ PRODUCTION CLOUD HOSTING & DOMAIN INFRASTRUCTURE
|      Domain: healthpulse-care.org | Status: [ ● ACTIVE / PROPAGATED ]
|
+--- 1. Authoritative DNS Resource Table -----+
| | TYPE | NAME / HOST | VALUE / DESTINATION | TTL | STATUS / HEALTH | |
| |-----+-----+-----+-----+-----+-----+ |
| | [ A ] | @ | 76.76.21.21 (Vercel Anycast Edge) | 3600s | ● Active & Propagated | |
| | [CNAME] | www | cname.vercel-dns.com | 3600s | ● Active & Propagated | |
| | [ MX ] | @ | 10 mail.healthpulse-care.org | 3600s | ● Priority 10 Active | |
| | [ TXT ] | @ | v=spf1 include:_spf.google.com ~all | 3600s | ● SPF Pass Enforced | |
| |-----+-----+-----+-----+-----+-----+ |
|
+--- 2. Interactive DNS Query Console (Live dig Simulator) -----+
| | Select Record: [ A Record (IPv4 Address) v ] [ 🔍 Execute DNS Query ] | |
| |
| | Query Output:
| | ;; ANSWER SECTION:
| | healthpulse-care.org. 3600 IN A 76.76.21.21
| | ;; SERVER: 1.1.1.1#53(Cloudflare Anycast Resolver)
| | ;; WHEN: Thu Oct 01 10:35:12 IST 2026
| |-----+-----+-----+-----+-----+-----+ |
|
+--- 3. Production Deployment Stepper -----+
| | [1] ICANN Domain Procurement (healthpulse-care.org via Cloudflare Registrar) | |
| | [2] Git Repository Connection (github.com/SmitroniX/SBL-EXPT) | |
| | [3] CI/CD Edge Deployment Triggered (Vercel Build Serverless Pipeline) | |
| | [4] Nameserver Delegation (ns1.vercel-dns.com / ns2.vercel-dns.com) | |
| | [5] Automatic SSL/TLS Certificate Provisioning (Let's Encrypt X.509 Wildcard) | |
| |-----+-----+-----+-----+-----+-----+ |
+-----+
```

3.3 Test Case Execution Results

Test Scenario	Action / Record Inspected	DNS Resolver Behavior	Observed Output Result	Status
Apex Resolution	Query healthpulse-care.org (A)	Matches Anycast IP	Resolves to 76.76.21.21 with TTL 3600s	PASS
Subdomain Alias	Query www.healthpulse-care.org	Follows CNAME pointer	Canonical name resolves to cname.vercel-dns.com	PASS

Test Scenario	Action / Record Inspected	DNS Resolver Behavior	Observed Output Result	Status
Mail Routing	Query <code>healthpulse-care.org</code> (MX)	Evaluates preference order	Returns <code>10 mail.healthpulse-care.org</code>	PASS
SPF Protection	Query TXT records	Verifies anti-spoof policy	Returns SPF record with Google Mail authorization	PASS
Edge Redirection	Inspect <code>netlify.toml</code> rules	Evaluates SPA catch-all	<code>/*</code> redirects seamlessly to <code>/index.html</code> with status 200	PASS
Security Headers	Inspect <code>vercel.json</code> headers	Evaluates HTTP headers	HSTS and <code>X-Frame-Options: SAMEORIGIN</code> applied	PASS

4. Conclusion & Verification

Experiment 09 provided a complete technical roadmap for cloud hosting. Through DNS record configuration, nameserver delegation, automated continuous integration, and Anycast content delivery network (CDN) routing, `healthpulse-care.org` demonstrated enterprise cloud readiness.